

M.A.M. SCHOOL OF ENGINEERING

(An Autonomous Institution)

Accredited by NAAC || Approved by AICTE, New Delhi || Affiliated to Anna University, Chennai
Trichy-Chennai Trunk Road, Siruganur, Tiruchirappalli -621 105.
TAMIL NADU, INDIA



CURRICULUM

REGULATION – 2024

(CHOICE BASED CREDIT SYSTEM AND OUTCOME BASED EDUCATION)

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

(ENGLISH MEDIUM)

I TO VIII SEMESTERS CURRICULUM

(Applicable for the students admitted from 2024-2025 onwards)

M.A.M. SCHOOL OF ENGINEERING

An Autonomous Institute

REGULATIONS 2024

CHOICE BASED CREDIT SYSTEM

B.E ELECTRONICS AND COMMUNICATION ENGINEERING

PROGRAMME EDUCATIONAL OBJECTIVES (PEO's)

- 1.To provide the students with a strong foundation in the required sciences in order to pursue studies in Electronics and Communication Engineering.
- 2.To gain adequate knowledge to become good professional in electronic and communication engineering associated industries, higher education and research.
- 3.To develop attitude in lifelong learning, applying and adapting new ideas and technologies as their field evolves.
- 4.To prepare students to critically analyze existing literature in an area of specialization and ethically develop innovative and research-oriented methodologies to solve the problems identified.
- 5.To inculcate in the students a professional and ethical attitude and an ability to visualize the engineering issues in a broader social context.

PROGRAM OUTCOMES (PO's)

PO1	Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7	Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and Team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs):

PSO1	Graduates will be able to apply the concepts of Electronics and Communication Engineering in the field of Embedded and VLSI Systems, IoT, Artificial Intelligence, Machine Learning, Communication and Networking.
PSO2	Graduates will be able to use the cutting edge software and hardware tools for analyzing, designing and implementing products.
PSO3	Graduates will be able to adapt to emerging electronics and communication technologies and develop innovative solutions for existing and newer problems.

CURRICULUM

M.A.M. SCHOOL OF ENGINEERING
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
REGULATIONS 2024
CHOICE BASED CREDIT SYSTEM
I TO VIII SEMESTERS CURRICULUM

Induction Program (Mandatory)	3 weeks duration
Induction program for students to be offered right at the start of the first year	- Physical activity , - Creative Arts, - Universal Human Values Literary, - Proficiency Modules, - Lectures by Eminent People Visits to local Areas, - Familiarization to Dept. / Branch & Innovations

B.E. ELECTRONICS AND COMMUNICATION ENGINEERING

SEMESTER I

S.No	Code No	Course	L	T	P	Credit	Maximum Marks			Cat.
							CA	ES	Total	
Theory Courses										
1	24HS101	Communicative English	3	0	0	3	40	60	100	HS
2	24HS102	Heritage of Tamil	1	0	0	1	40	60	100	HS
3	24BS101	Matrices & Calculus	3	1	0	4	40	60	100	BS
4	24ES101	Problem solving and Python Programming	3	0	0	3	40	60	100	ES
Theory courses with Laboratory components										
5	24BS102	Engineering Chemistry	3	0	2	4	50	50	100	BS
Laboratory courses										
6	24HS103	Communicative English Laboratory	0	0	2	1	60	40	100	HS
7	24ES103	Problem solving and Python Programming Laboratory	0	0	4	2	60	40	100	ES
8	24ES106	Engineering Practices Laboratory	0	0	2	1	60	40	100	ES
9	24ES107	Workshop Practices Laboratory	0	0	2	1	60	40	100	ES
TOTAL			13	1	12	20				

SEMESTER II

S.No	Code No	Course	L	T	P	Credit	Maximum Marks			Cat.
							CA	ES	Total	
Theory Courses										
1	24HS201	தமிழரும் தொழில் துட்பமும் / Tamils and Technology	1	0	0	1	40	60	100	HS
2	24HS202	Professional English	2	0	0	2	40	60	100	HS
3	24BS201	Transforms and Partial Differential Equations	3	1	0	4	40	60	100	BS
4	24ES201	Design Thinking	2	0	0	2	40	60	100	ES
Theory courses with Laboratory components										
5	24BS204	Physics for Engineers	3	0	2	4	50	50	100	BS
6	24ES202	Electronic Circuit Analysis	2	1	2	4	50	50	100	ES
7	24ES206	C Programming	3	0	2	4	50	50	100	ES
Laboratory courses										
8	24ES205	Engineering Drawing	0	0	4	2	60	40	100	ES
9	24TP201	Aptitude Skills-I and Communication skills – I	0	0	2	1	100	-	100	EEC
TOTAL			16	2	12	24				

SEMESTER III

S.No	Code No	Course	L	T	P	Credit	Maximum Marks			Cat.
							CA	ES	Total	
Theory Courses										
1	24BS301	Statistics and Numerical Method	3	1	0	4	40	60	100	BS
2	24EC301	Digital System Design	3	0	0	3	40	60	100	PC
3	24CS302	Object Oriented Programming	3	0	0	3	40	60	100	PC
4	24EEE02	Control System	3	0	0	3	40	60	100	PC
Theory courses with Laboratory components										
5	24EC302	Electronic Devices and Circuits	3	0	2	4	50	50	100	PC
6	24EC303	Signals Processing	3	0	2	4	40	60	100	PC
Laboratory courses										
7	24CS304	Object Oriented Programming Lab	0	0	4	2	60	40	100	PC
8	24EC304	Digital System Design Lab	0	0	4	2	60	40	100	PC
9	24TP301	Aptitude Skills-II and Communication Skills II	0	0	2	1	100	-	100	EEC
TOTAL			18	1	14	26				

SEMESTER IV

S.No	Code No	Course	L	T	P	Credit	Maximum Marks			Cat.
							CA	ES	Total	
Theory courses										
1	24EC401	Electromagnetic Fields	3	0	0	3	40	60	100	PC
2	24EC402	Communication Systems	3	0	0	3	40	60	100	PC
3	24EC403	Linear Integrated Circuits	3	0	0	3	40	60	100	PC
4	24MC401	Environmental Sciences	3	0	0	0	40	60	100	MC
5	24MC402	Disaster Management and preparedness	2	0	0	2	40	60	100	MC
Theory courses with Laboratory components										
6	24EC404	Networks and Security	3	0	2	4	50	50	100	PC
7	24EC405	Microprocessors and DSP architecture	3	0	2	4	50	50	100	PC
Laboratory courses										
8	24EC406	Communication Systems Lab	0	0	4	2	60	40	100	PC
9	24EC407	Linear Integrated Circuits Lab	0	0	4	2	60	40	100	PC
10	24TP401	Aptitude Skills- III & Technical Skills I	0	0	2	1	100	-	100	EEC
TOTAL			18	0	14	24				

SEMESTER V

S.No	Code No	Course	L	T	P	Credit	Maximum Marks			Cat.
							CA	ES	Total	
Theory courses										
1	24EC501	Transmission Lines and RF systems	3	0	0	3	40	60	100	PC
2	24EC502	VLSI and Chip Design	3	0	0	3	40	60	100	PC
3		Professional Elective-I	3	0	0	3	40	60	100	PE
4		Professional Elective-II	2	0	2	3	40	60	100	PE
5		Open Elective-I	3	0	0	3	40	60	100	OE
Theory courses with Laboratory components										
6	24EC503	Embedded System and IOT	3	0	2	4	50	50	100	PC
Laboratory courses										
7	24EC504	VLSI and Chip Design Lab	0	0	4	2	60	40	100	PC
8	24TP501	Aptitude Skills- IV & Technical Skills II	0	0	2	1	100	-	100	EEC
TOTAL			18	0	8	22				

SEMESTER VI

S. No	Code No	Course	L	T	P	Credit	Maximum Marks			Cat.
							CA	ES	Total	
Theory courses										
1	24HS601	Principles of Management	3	0	0	3	40	60	100	HS
2		Professional Elective-III	3	0	0	3	40	60	100	PE
3		Professional Elective-IV	2	0	2	3	50	50	100	PE
4		Open Elective-II	3	0	0	3	40	60	100	OE
Theory courses with Laboratory components										
5	24EC601	Wireless Communication	3	0	2	4	50	50	100	PC
6	24EC602	Artificial Intelligence and Machine Learning	3	0	2	4	50	50	100	PC
Laboratory courses										
7	24TP601	Aptitude Skills V & Technical Skills III	0	0	2	1	100	-	100	EEC
TOTAL			17	0	12	21				

SEMESTER VII

S.No	Code No	Course	L	T	P	Credit	Maximum Marks			Cat.
							CA	ES	Total	
Theory courses										
1	24HS701	Human Values and Ethics	3	0	0	3	40	60	100	HS
2	24EC701	Antenna and Microwave Engineering	3	0	0	3	40	60	100	PC
3		Professional Elective-V	3	0	0	3	40	60	100	PE
4		Open Elective-III	3	0	0	3	40	60	100	OE
Laboratory courses										
5	24EC702	Mini Project	0	0	4	2	60	40	100	EEC
6	24EC703	Internship	0	0	0	2	100	-	100	EEC
TOTAL			12	0	0	16				

SEMESTER VIII

S. No	Code No	Course	L	T	P	Credit	Maximum Marks			Cat.
							CA	ES	Total	
Laboratory course										
1	24EC801	Project Work	0	0	20	10	60	40	100	EEC
TOTAL			0	0	20	10				

PROFESSIONAL ELECTIVE - I
COMMUNICATION NETWORKS

S. No	Code No	Course	L	T	P	C	Maximum Marks			Cat.
							CA	ES	Total	
1	24ECX01	Wireless Sensor Networks	3	0	0	3	40	60	100	PE
2	24ECX02	Wireless Broad Band Networks	3	0	0	3	40	60	100	PE
3	24ECX03	Software Defined Networks	3	0	0	3	40	60	100	PE
4	24ECX04	Massive MIMO Networks	3	0	0	3	40	60	100	PE
5	24ECX05	Optical Communication Networks	3	0	0	3	40	60	100	PE
6	24ECX06	4G/5G Communication Networks	3	0	0	3	40	60	100	PE
7	24ECX07	Mobile Communication and Networks	3	0	0	3	40	60	100	PE
8	24ECX08	Adhoc and sensor networks	3	0	0	3	40	60	100	PE

PROFESSIONAL ELECTIVE – II
SPACE TECHNOLOGIES

S. No	Code No	Course	L	T	P	C	Maximum Marks			Cat.
							CA	ES	Total	
1	24ECX09	Radar Technologies	3	0	0	3	40	60	100	PE
2	24ECX10	Avionics Systems	3	0	0	3	40	60	100	PE
3	24ECX11	Positioning and Navigation Systems	3	0	0	3	40	60	100	PE
4	24ECX12	Satellite Communication	3	0	0	3	40	60	100	PE
5	24ECX13	Remote Sensing	3	0	0	3	40	60	100	PE
6	24ECX14	Rocketry and Space Mechanics	3	0	0	3	40	60	100	PE
7	24ECX15	Rocket Propulsion	3	0	0	3	40	60	100	PE
8	24ECX16	Space Engineering	3	0	0	3	40	60	100	PE

PROFESSIONAL ELECTIVE - III

SIGNAL PROCESSING & ADVANCED COMMUNICATION

S. No	Code No	Course	L	T	P	C	Maximum Marks			Cat.
							CA	ES	Total	
1	24ECX17	Image Processing	3	0	0	3	40	60	100	PE
2	24ECX18	Speech Processing	3	0	0	3	40	60	100	PE
3	24ECX19	Advanced Digital Signal Processing	3	0	0	3	40	60	100	PE
4	24ECX20	Software Defined Radio	3	0	0	3	40	60	100	PE
5	24ECX21	Computer Vision	3	0	0	3	40	60	100	PE
6	24ECX22	Signal Integrity	3	0	0	3	40	60	100	PE
7	24ECX23	Underwater Communication	3	0	0	3	40	60	100	PE
8	24ECX24	Advanced wireless communication techniques	3	0	0	3	40	60	100	PE

PROFESSIONAL ELECTIVE – IV

SEMICONDUCTOR CHIP DESIGN

S. No	Code No	Course	L	T	P	C	Maximum Marks			Cat.
							CA	ES	Total	
1	24ECX25	Low Power IC Design	3	0	0	3	40	60	100	PE
2	24ECX26	VLSI Testing and Design For Testability	3	0	0	3	40	60	100	PE
3	24ECX27	Network on chip	3	0	0	3	40	60	100	PE
4	24ECX28	Semiconductor Packaging	3	0	0	3	40	60	100	PE
5	24ECX29	System on chip design	3	0	0	3	40	60	100	PE
6	24ECX30	DSP architecture and programming	3	0	0	3	40	60	100	PE
7	24ECX31	Mixed signal IC testing	3	0	0	3	40	60	100	PE
8	24ECX32	IOT processors	3	0	0	3	40	60	100	PE

PROFESSIONAL ELECTIVE – V

BIO MEDICAL TECHNOLOGIES

S.No	Code No	Course	L	T	P	C	Maximum Marks			Cat.
							CA	ES	Total	
1	24ECX33	Wearable Devices	3	0	0	3	40	60	100	PE
2	24ECX34	Therapeutic Equipment	3	0	0	3	40	60	100	PE
3	24ECX35	Human Assist Devices	3	0	0	3	40	60	100	PE
4	24ECX36	Medical Imaging Systems	3	0	0	3	40	60	100	PE
5	24ECX37	Body Area Networks	3	0	0	3	40	60	100	PE
6	24ECX38	Brain Computer Interface and Applications	3	0	0	3	40	60	100	PE
7	24ECX39	Fundamentals of Bio science	3	0	0	3	40	60	100	PE
8	24ECX40	Critical Care Equipment	3	0	0	3	40	60	100	PE

OPEN ELECTIVES										
S.No	Code No	Course	L	T	P	C	Maximum Marks			Cat.
							CA	ES	Total	
1	24ECY01	Wireless technologies	3	0	0	3	40	60	100	OE
2	24ECY02	Principles of Modern Communication systems	3	0	0	3	40	60	100	OE
3	24ECY03	Introduction to Image Processing	3	0	0	3	40	60	100	OE
4	24ECY04	Sensors and Transducers	3	0	0	3	40	60	100	OE
5	24ECY05	Fundamentals of Speech Processing	3	0	0	3	40	60	100	OE
6	24ECY06	Introduction to IOT processors	3	0	0	3	40	60	100	OE
7	24ECY07	Project Report Writing	3	0	0	3	40	60	100	OE
8	24ECY08	Entrepreneurship Development	3	0	0	3	40	60	100	OE
9	24ECY09	Drinking Water Supply and treatment	3	0	0	3	40	60	100	OE
10	24ECY10	Drone technologies	3	0	0	3	40	60	100	OE
11	24ECY11	Urban Agriculture	3	0	0	3	40	60	100	OE

Summary of Credit Distribution

S.No	Category	Credits Per Semester	Total	AICTE	AU
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		I	II	III	IV	V	VI	VII	VIII	Credit		
1	HS	5	3				3	3		14	15	12
2	BS	8	8	4						20	23	25
3	ES	7	12							19	17	21
4	PC			21	21	12	8	3		65	61	58
5	PE					6	6	3		15	12	18
6	OE					3	3	3		9	12	12
7	MC				2					2		
8	EEC		1	1	1	1	1	4	10	19	20	16
TOTAL		20	24	26	24	22	21	16	10	20	163	

- BS - Basic Science
- ES - Engineering Sciences
- HS - Humanities and Social Sciences
- PC - Professional Core
- PE - Professional Elective
- OE - Open Elective
- EEC - Employability Enhancement Course
- MC - Mandatory Course
- CA - Continuous Assessment
- ES - End Semester Examination